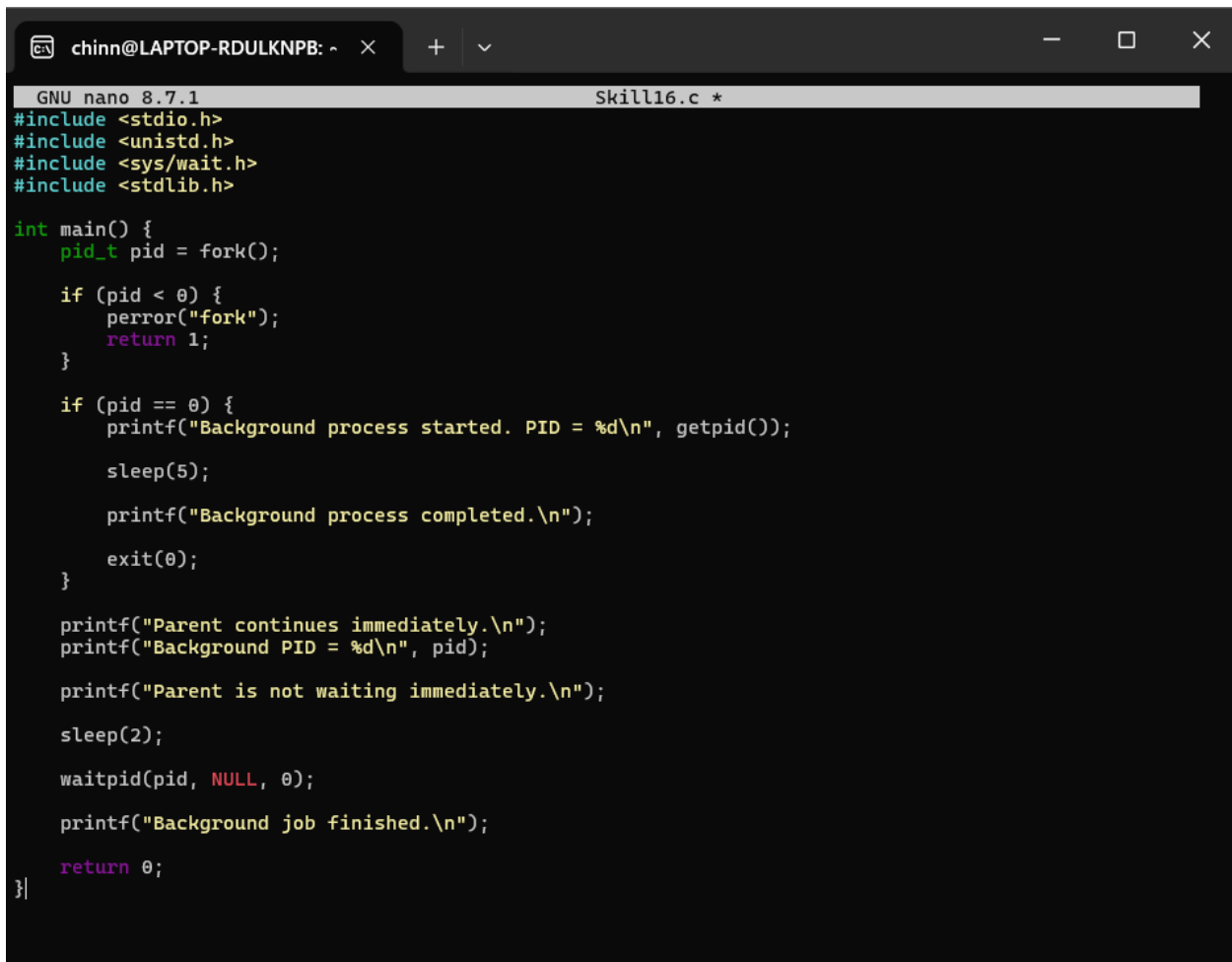


Name: V.Siri Chandana
Roll No:2520030609
Section:2
Subject:OS

Skill Programs

16.



```
GNU nano 8.7.1 skill16.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Background process started. PID = %d\n", getpid());

        sleep(5);

        printf("Background process completed.\n");

        exit(0);
    }

    printf("Parent continues immediately.\n");
    printf("Background PID = %d\n", pid);

    printf("Parent is not waiting immediately.\n");

    sleep(2);

    waitpid(pid, NULL, 0);

    printf("Background job finished.\n");

    return 0;
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano Skill16.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc Skill16.c -o Skill16
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./Skill16
Parent continues immediately.
Background PID = 1287
Parent is not waiting immediately.
Background process started. PID = 1287
Background process completed.
Background job finished.
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ |
```

17.

```
chinn@LAPTOP-RDULKNPB: ~ × + ▾
GNU nano 8.7.1 Skill17.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Background process started. PID = %d\n", getpid());

        sleep(5);

        printf("Background process completed.\n");
        exit(0);
    }

    printf("Parent continues immediately.\n");
    printf("Background PID = %d\n", pid);

    printf("Parent is not waiting immediately.\n");

    sleep(2);

    waitpid(pid, NULL, 0);

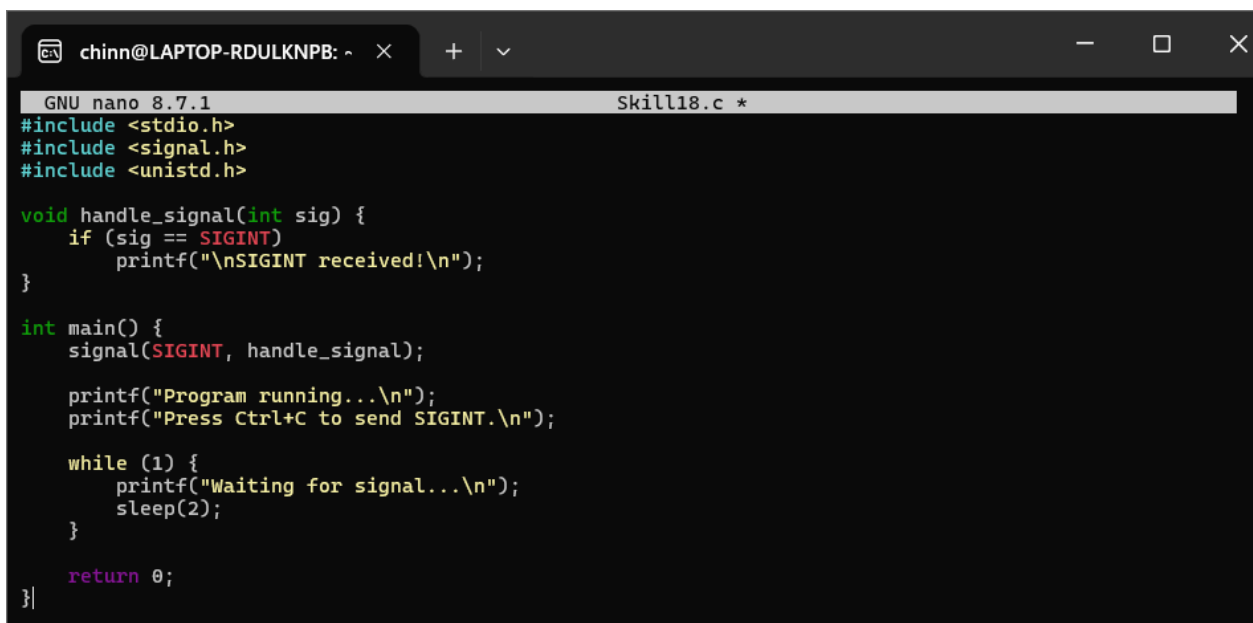
    printf("Background job finished.\n");

    return 0;
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano Skill17.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc Skill17.c -o Skill17
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./Skill17
Parent continues immediately.
Background PID = 1329
Parent is not waiting immediately.
Background process started. PID = 1329
Background process completed.
Background job finished.
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ |
```

18.



```
chinn@LAPTOP-RDULKNPB: ~ × + ▾ - □ ×
GNU nano 8.7.1 skill18.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handle_signal(int sig) {
    if (sig == SIGINT)
        printf("\nSIGINT received!\n");
}

int main() {
    signal(SIGINT, handle_signal);

    printf("Program running...\n");
    printf("Press Ctrl+C to send SIGINT.\n");

    while (1) {
        printf("Waiting for signal...\n");
        sleep(2);
    }

    return 0;
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano Skill18.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc Skill18.c -o Skill18
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./Skill18
Program running...
Press Ctrl+C to send SIGINT.
Waiting for signal...
Waiting for signal...
Waiting for signal...
Waiting for signal...
^C
SIGINT received!
Waiting for signal...
Waiting for signal...
Waiting for signal...
Waiting for signal...
Waiting for signal...
^Z
[1]+  Stopped                  ./Skill18
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$
```

19.

```
chinn@LAPTOP-RDULKNPB: ~ x + v - □ ×
GNU nano 8.7.1 Skill19.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handle_signal(int sig) {
    if (sig == SIGINT)
        printf("\nSIGINT received!\n");
}

int main() {
    signal(SIGINT, handle_signal);

    printf("Program running...\n");
    printf("Press Ctrl+C to send SIGINT.\n");

    while (1) {
        printf("Waiting for signal...\n");
        sleep(2);
    }

    return 0;
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano Skill19.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc Skill19.c -o Skill19
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./Skill19
Program running...
Press Ctrl+C to send SIGINT.
Waiting for signal...
Waiting for signal...
Waiting for signal...
Waiting for signal...
^C
SIGINT received!
Waiting for signal...
Waiting for signal...
Waiting for signal...
^Z
[2]+  Stopped                  ./Skill19
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ |
```

20.

```
chinn@LAPTOP-RDULKNPB: ~ × + ▾
GNU nano 8.7.1 Skill20.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <stdlib.h>

void sigint_handler(int sig) {
    printf("\nSIGINT caught by process.\n");
    printf("Cleaning resources...\n");
    printf("Program exiting safely.\n");

    exit(0);
}

int main() {
    signal(SIGINT, sigint_handler);

    printf("Process PID: %d\n", getpid());
    printf("Press Ctrl+C to terminate safely.\n");

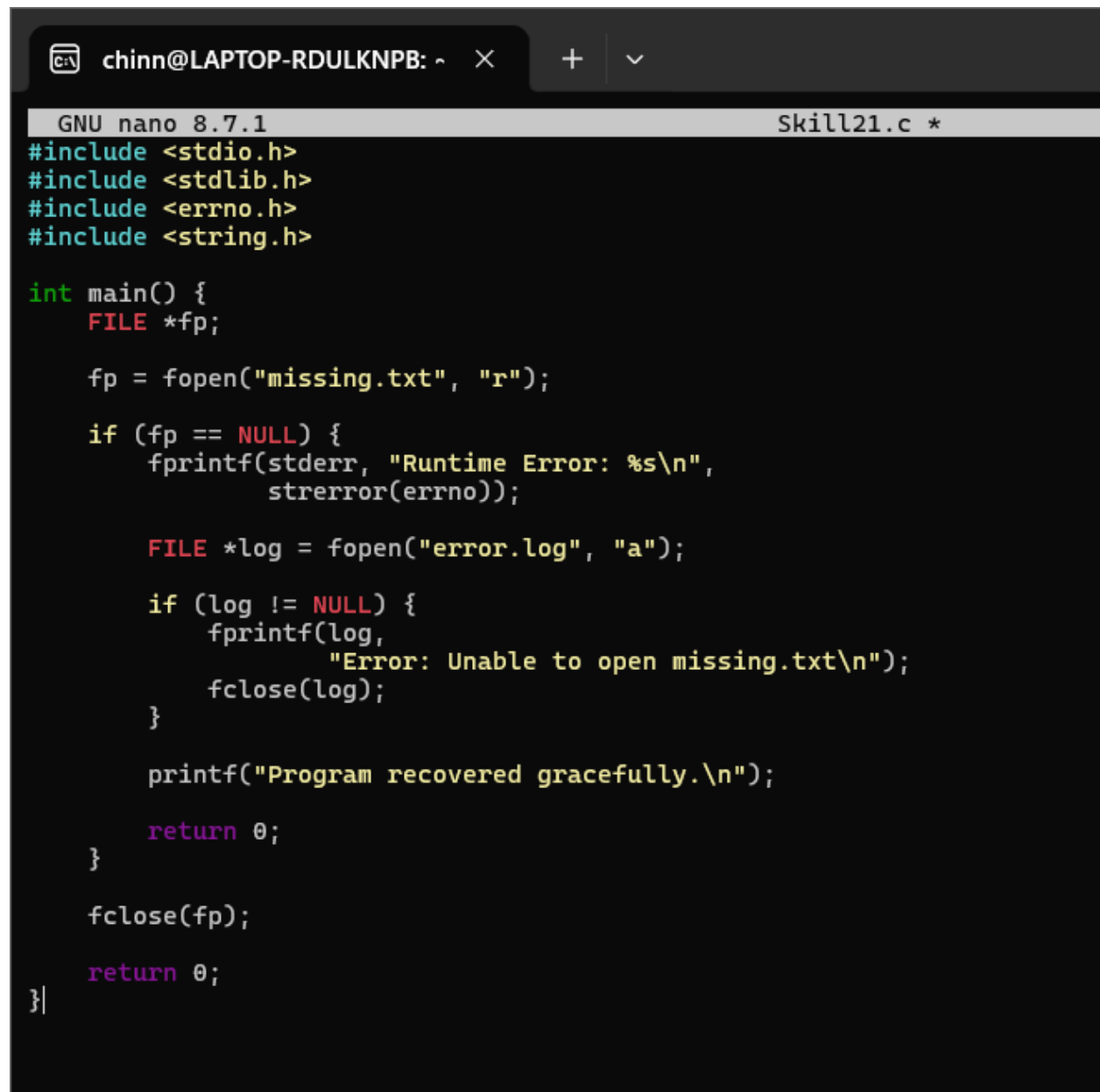
    while (1) {
        printf("Running...\n");
        sleep(2);
    }

    return 0;
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano Skill20.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc Skill20.c -o Skill20
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./Skill20
Process PID: 1457
Press Ctrl+C to terminate safely.
Running...
Running...
Running...
^C
SIGINT caught by process.
Cleaning resources...
Program exiting safely.
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ |
```

21.



The image shows a terminal window with a dark background. At the top, there is a tab labeled 'chinn@LAPTOP-RDULKNPB: ~' with a close button (X) and a dropdown arrow (v). Below the tab, the terminal header shows 'GNU nano 8.7.1' on the left and 'Skill21.c *' on the right. The main area contains C code for a program named 'Skill21.c'. The code includes standard headers, defines a main function, attempts to open 'missing.txt', handles the error by logging to 'error.log', and prints a recovery message.

```
chinn@LAPTOP-RDULKNPB: ~ × + v
GNU nano 8.7.1 Skill21.c *
#include <stdio.h>
#include <stdlib.h>
#include <errno.h>
#include <string.h>

int main() {
    FILE *fp;

    fp = fopen("missing.txt", "r");

    if (fp == NULL) {
        fprintf(stderr, "Runtime Error: %s\n",
                strerror(errno));

        FILE *log = fopen("error.log", "a");

        if (log != NULL) {
            fprintf(log,
                    "Error: Unable to open missing.txt\n");
            fclose(log);
        }

        printf("Program recovered gracefully.\n");

        return 0;
    }

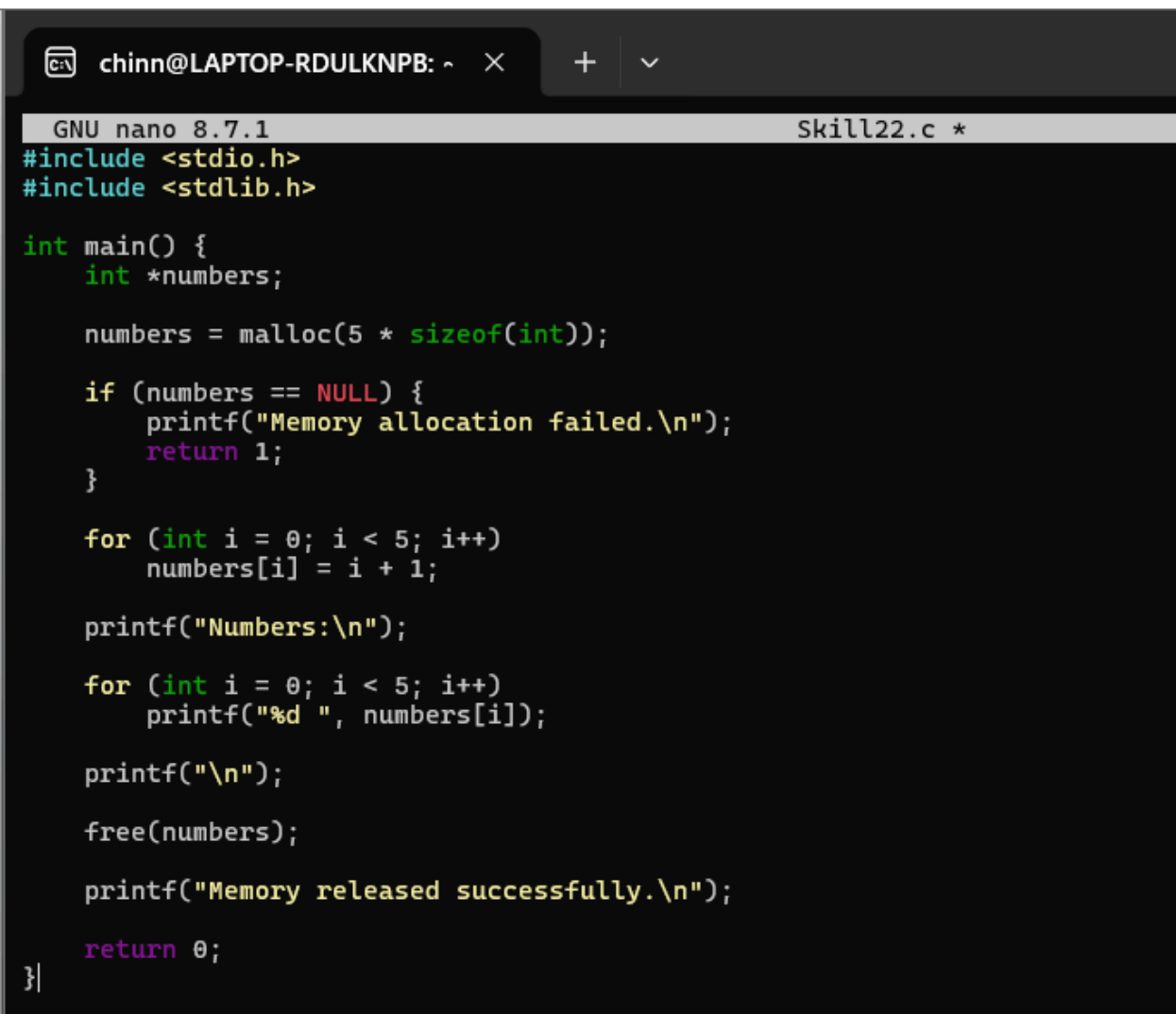
    fclose(fp);

    return 0;
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano Skill21.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc Skill21.c -o Skill21
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./Skill21
Runtime Error: No such file or directory
Program recovered gracefully.
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ |
```

22.



```
chinn@LAPTOP-RDULKNPB: ~ × + ▾
GNU nano 8.7.1 Skill22.c *
#include <stdio.h>
#include <stdlib.h>

int main() {
    int *numbers;

    numbers = malloc(5 * sizeof(int));

    if (numbers == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }

    for (int i = 0; i < 5; i++)
        numbers[i] = i + 1;

    printf("Numbers:\n");

    for (int i = 0; i < 5; i++)
        printf("%d ", numbers[i]);

    printf("\n");

    free(numbers);

    printf("Memory released successfully.\n");

    return 0;
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano Skill22.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc Skill22.c -o Skill22
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./Skill22
Numbers:
1 2 3 4 5
Memory released successfully.
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ |
```

23.

```
chinn@LAPTOP-RDULKNPB: ~ × + ▾
GNU nano 8.7.1 Skill23.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    int pipe1[2];
    int pipe2[2];

    pipe(pipe1);
    pipe(pipe2);

    pid_t p1 = fork();

    if (p1 == 0) {
        dup2(pipe1[1], STDOUT_FILENO);

        close(pipe1[0]);
        close(pipe1[1]);
        close(pipe2[0]);
        close(pipe2[1]);

        execlp("ls", "ls", NULL);

        perror("ls");
        exit(1);
    }

    pid_t p2 = fork();

    if (p2 == 0) {
        dup2(pipe1[0], STDIN_FILENO);
        dup2(pipe2[1], STDOUT_FILENO);

        close(pipe1[0]);
        close(pipe1[1]);
        close(pipe2[0]);
        close(pipe2[1]);

        execlp("grep", "grep", ".c", NULL);

        perror("grep");
        exit(1);
    }

    pid_t p3 = fork();

    if (p3 == 0) {
        dup2(pipe2[0], STDIN_FILENO);

        close(pipe1[0]);
        close(pipe1[1]);
        close(pipe2[0]);
        close(pipe2[1]);

        execlp("wc", "wc", "-l", NULL);

        perror("wc");
        exit(1);
    }

    close(pipe1[0]);
    close(pipe1[1]);
    close(pipe2[0]);
    close(pipe2[1]);

    waitpid(p1, NULL, 0);
    waitpid(p2, NULL, 0);
    waitpid(p3, NULL, 0);

    printf("Pipeline completed successfully.\n");

    return 0;
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano Skill23.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc Skill23.c -o Skill23
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./Skill23
32
Pipeline completed successfully.
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ |
```

24.

```
GNU nano 8.7.1 Skill24.c *
#include <stdio.h>

int main() {
    printf("=====\n");
    printf("          PROJECT ORGANIZATION\n");
    printf("=====\n");

    printf("1. Organize Source Code\n");
    printf("2. Improve Build Scripts\n");
    printf("3. Add README\n");
    printf("4. Verify Repository Structure\n");
    printf("5. Tag Releases\n");
    printf("6. Validate Deliverables\n");

    printf("\n=====\n");
    printf("          DEMO SCENARIOS\n");
    printf("=====\n");

    printf("1. Create Demo Scenarios\n");
    printf("2. Validate Features\n");
    printf("3. Rehearse Presentation\n");
    printf("4. Demonstrate Pipelines\n");
    printf("5. Demonstrate Signals\n");
    printf("6. Final Project Review\n");

    printf("\nProject preparation completed successfully.\n");

    return 0;
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano Skill24.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc Skill24.c -o Skill24
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./Skill24
=====
                PROJECT ORGANIZATION
=====
1. Organize Source Code
2. Improve Build Scripts
3. Add README
4. Verify Repository Structure
5. Tag Releases
6. Validate Deliverables

=====
                DEMO SCENARIOS
=====
1. Create Demo Scenarios
2. Validate Features
3. Rehearse Presentation
4. Demonstrate Pipelines
5. Demonstrate Signals
6. Final Project Review

Project preparation completed successfully.
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ |
```